...rollment Number)

Enrollment No. 12701182013

MID-TERM EXAMINATION

B. Tech. ECE-AI Semester: III

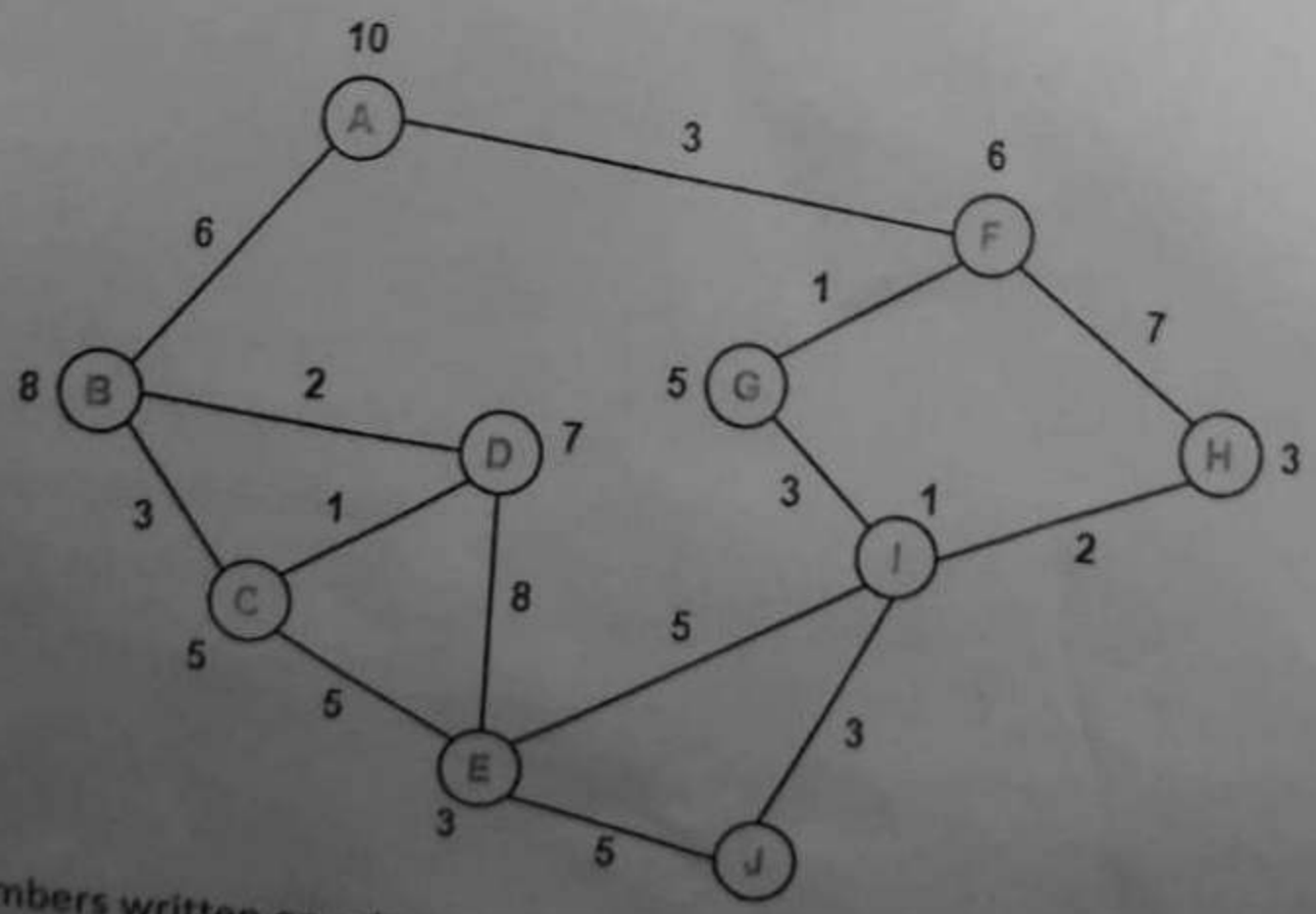
(October 2024) OFF LINE mode

Subject Code: BAI 205 | **Neural Networks and Artificial Intelligence**

Time : 1 ½Hours | **Maximum Marks : 30**

Note: Q. 1 is compulsory.

Q1			
	(a) Define intelligent agents, simplex reflex agents and goal-based agents?	(2.5*4)	CO
	(b) Explain the approaches to AI and contribution of AI in various fields?		CO1
	(c) Define heuristic? Explain any two commonly used heuristic functions in AI?		CO1
	(d) Compare informed and uninformed search in a tabular form? State any two examples of uninformed AI and informed AI search techniques?		CO2
Q2	**(Attempt any Two Parts) UNIT-1 (CO1)**		CO2
	a) Explain PEAS and properties of task environments. Write the PEAS description of the task environment for an automated car driving system.		**(5,5)**
	b) State task domains of AI? Explain the Machine Learning and Natural Language Processing domains of AI?		CO1
	c) Explain the state space representation of AI problem? Define the water jug problem in AI, and suggest a solution of it?		CO1
Q3	**(Attempt any Two Parts) UNIT-2 (CO2)**		CO1
	a) Discuss hill climbing in AI and explain all the drawbacks in detail?		**(5,5)**
	b) Explain alpha-beta pruning? How can it improve the Minmax algorithm?		CO2
	c) Explain the A* algorithm in AI with advantages & disadvantages?		CO2
	Consider the following graph-		CO2



The numbers written on edges represent the distance between the nodes. The numbers written on nodes represent the heuristic value. Find the most cost-effective path to reach from start state A to final state J using A* Algorithm?